Divisorial Contraction in the Kähler MMP

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Contents

Big picture

Once the cone theorem produces a $(K_X + B)$-negative extremal ray R, the next step of the minimal model program is to *contract* it. This note treats the case where R is of **divisorial type** — equivalently $\dim \mathrm{Null}(\alpha) = 2$ for the nef and big supporting class α — for a $\mathbb{Q}$-factorial compact Kähler 3-fold klt pair. Two proofs of the existence of the divisorial contraction are presented and compared.

?? fixes the definition, records the structural properties any divisorial contraction must have, and states the theorem both proofs establish. **??** is the Das–Hacon approach: an all-dimensional PLT contraction theorem (**??**) supplies the engine, and the contraction is manufactured by lifting to a $\mathbb{Q}$-factorial dlt model, running an α'-trivial MMP that surgers only the null locus, and showing the resulting map descends to a base-point-free morphism. **??** is the earlier Höring–Peternell approach, tied to surface geometry: the null locus S is contracted to a point (nef dimension 0) or fibred over a curve and contracted by Nakano's theorem plus a divisorial extension theorem (nef dimension 1). **??** compares the two.

The references are [**?**] for the Das–Hacon approach, [**?**] and [**?**] for the Höring–Peternell approach. Throughout, (X, B) is a compact Kähler pair, $\alpha \in H^{1,1}_{\mathrm{BC}}(X)$ is a Bott–Chern class, $\overline{\mathrm{NA}}(X)$ is the generalized Mori cone (the closure of the cone of classes of positive currents), and $\mathrm{Null}(\alpha) = \bigcup_{\int_V (\alpha|_V)^{\dim V} = 0} V$ is the null locus of a nef and big class.

1 Setup: divisorial contractions in the Kähler MMP

Definition 1.1 ([?, Def. 3.10.4])**.** Let $(X, B + \boldsymbol{\beta})$ be a $\mathbb{Q}$-factorial compact Kähler pair. A bimeromorphic morphism $f\colon X \to Z$ to a normal compact complex space is a **divisorial contraction** if

(1) the relative Picard number is $\rho(X/Z) = 1$ and the exceptional locus $\mathrm{Ex}(f)$ is a divisor, and

(2) $-(K_X + B + \boldsymbol{\beta})$ is f-ample (equivalently, relatively Kähler over Z).

Any f meeting **??** inherits the following, exactly as in the projective MMP; these are the properties the two existence proofs are engineered to realize.

Proposition 1.2 (Basic properties, [?, Thm. 2.5.1]). *Let $f : X \longrightarrow Z$ be a divisorial contraction of a $(K_X + B)$-negative extremal ray, with X $\mathbb{Q}$-factorial and DLT. Then:*

(1) $\mathrm{Ex}(f) = E$ *is a single* prime *divisor;*

(2) $\rho(X/Z) = 1$, *so* $\rho(Z) = \rho(X) - 1$;

(3) Z *is* $\mathbb{Q}$*-factorial;*

(4) $(Z, B_Z = f_*B)$ *is again DLT (resp. KLT), and one may write* $K_X + B = f^*(K_Z + B_Z) + eE$ *with discrepancy* $e > 0$.

The theorem toward which everything below builds is the following.

Theorem 1.3 ([?, Thm. 6.9]). *Let (X, B) be a strongly $\mathbb{Q}$-factorial compact Kähler 3-fold klt pair such that*

(1) $K_X + B$ *is pseudo-effective,*

(2) $\alpha = [K_X + B + \beta]$ *is nef and big for some Kähler class* β, *and*

(3) $R = \alpha^{\perp} \cap \overline{\mathrm{NA}}(X)$ *is an extremal ray of* ***divisorial type****, i.e.* $\dim \mathrm{Null}(\alpha) = 2$.

Then the divisorial contraction $f : X \longrightarrow Z$ of R exists, and there is a Kähler class α_Z on Z with $\alpha = f^*\alpha_Z$.

Note that since R is α-trivial and β is Kähler, condition (3) forces R to be $(K_X + B)$-negative: $(K_X + B) \cdot R = (\alpha - \beta) \cdot R = -\beta \cdot R < 0$. The class α plays a double role — it is the supporting class cutting out R, and the target of a descent $\alpha = f^*\alpha_Z$ that certifies base-point freeness.

2 The Das–Hacon approach

The strategy is dimension-independent in its engine and surgical in its execution. The engine is a contraction theorem for PLT pairs (**??**). Its input — a divisor S carrying a fibration with the right relative positivity — is produced by localizing on the null locus (**??**), passing to a dlt model (**??**), and running an MMP that touches only that locus (**????**). The output is shown to be a genuine base-point-free morphism (**????**).

2.1 The engine: a contraction theorem for analytic PLT pairs

Theorem 2.1 ([?, Thm. 5.8]). *Let $(X, S + B + \boldsymbol{\beta})$ be a generalized PLT pair with X a compact analytic variety and S $\mathbb{Q}$-Cartier. Suppose*

(1) $\lfloor S + B \rfloor = S$ *is irreducible,*

(2) there is a proper morphism $\pi\colon S \to T$ with $\pi_*\mathcal{O}_S = \mathcal{O}_T$,

(3) $-(K_X + S + B + \boldsymbol{\beta}_X)|_S$ is relatively Kähler over T, and

(4) $-S|_S$ is π-ample.

Then there is a bimeromorphic morphism $p\colon X \to Z$ with $p|_S = \pi$ and $p|_{X\setminus S}$ an isomorphism. If p is of flipping type, the flip exists.

Idea of proof. Apply Fujiki's blow-down theorem, but not to S directly — S is only $\mathbb{Q}$-Cartier. Instead blow down a Cartier thickening $A = \ell S$. Two things must be arranged: that $-A|_A$ stays π_A-ample on the thickening (an "ample on the reduction $\Rightarrow$ ample" argument), and, the technical heart, the vanishing $R^1\pi_*\mathcal{E}_k = 0$ of the first direct image of the graded pieces $\mathcal{E}_k = \mathcal{O}_X(-kS)/\mathcal{O}_X(-(k+1)S)$. This vanishing is what lets the ring structure of $\bigoplus \pi_*\mathcal{O}_{S_k}$ integrate the infinitesimal fibrations $\pi_k\colon S_k \to T_k$ into an honest contraction. It follows from Kawamata–Viehweg vanishing applied to the GKLT pair extracted from $(X, S + B + \boldsymbol{\beta})$ by adjunction, since $-(K_X + S + B + \boldsymbol{\beta}_X)|_S$ is relatively big and nef.

Sketch. By adjunction S is normal, $(S, B_S + \boldsymbol{\beta}_S)$ is GKLT and $-(K_S + B_S + \boldsymbol{\beta}_S) = -(K_X + S + B + \boldsymbol{\beta}_X)|_S$ is Kähler over T. Choosing $\mathcal{L} = \mathcal{O}_X(-K_X - (k+1)S)$ in the adjunction sequence yields

$$0 \to \mathcal{O}_X(-(k+1)S) \to \mathcal{O}_X(-kS) \to \mathcal{E}_k \to 0,$$

with $\mathcal{E}_k$ a rank-1 reflexive sheaf on S; writing S_k for the thickening $\mathcal{O}_{S_k} = \mathcal{O}_X/\mathcal{O}_X(-kS)$ gives the ring sequence $0 \to \mathcal{E}_k \to \mathcal{O}_{S_{k+1}} \to \mathcal{O}_{S_k} \to 0$.
Vanishing. Locally over T one has $\mathcal{E}_k = \mathcal{O}_S(G_k)$ with

$$G_k \sim_{\mathbb{Q}} K_S + (B_S - \Delta_k) + \boldsymbol{\beta}_S - (K_S + B_S + \boldsymbol{\beta}_S) - kS|_S, \qquad 0 \le \Delta_k \le B_S,$$

so $(S, (B_S - \Delta_k) + \boldsymbol{\beta}_S)$ is GKLT and the twist $-(K_S + B_S + \boldsymbol{\beta}_S) - kS|_S$ is relatively Kähler; Kawamata–Viehweg vanishing gives $R^1\pi_*\mathcal{E}_k = 0$ (when B_S is not $\mathbb{Q}$-Cartier one first passes to a small $\mathbb{Q}$-factorialization $q\colon \tilde{S} \to S$, applies vanishing there, and descends by a spectral-sequence argument). Hence $\pi_*\mathcal{O}_{S_{k+1}} \to \pi_*\mathcal{O}_{S_k}$ is surjective, so $T_k := \mathrm{Specan}(\pi_*\mathcal{O}_{S_k})$ are compatible thickenings of T and the $\pi_k\colon S_k \to T_k$ are morphisms of non-reduced spaces.
Blow-down. Fix ℓ with $A = \ell S$ Cartier. Since $-A|_A$ pulls back to the π-ample $-\ell S|_S$ on the reduction and the induced map on bases is finite, $-A|_A$ is π_A-ample; a change-of-index plus Serre vanishing argument upgrades this to $R^i\pi_{A,*}\mathcal{O}_A(-kA) = 0$ for $i, k \ge 1$. Fujiki's blow-down theorem then produces $p\colon X \to Z$ with $p|_A = \pi_A$ and $p|_{X\setminus A}$ an isomorphism; restricting, $p|_S = \pi$. Flips of flipping type exist by analytic BCHM [**?**]. ■

Remark 2.2. The Cartier hypothesis on S is exactly what would let one quote KV vanishing directly; its failure is absorbed into the "variation of the different" correction term Δ_k, whose GKLT-ness is what the proof buys. This is why the theorem holds *in all dimensions* and is the load-bearing input to the 3-fold construction below.

2.2 Step 1: the null locus is an irreducible Moishezon surface

Proposition 2.3 ([?, Prop. 6.10]). *Under the hypotheses of* **??** *there is an irreducible divisor S with* $\mathrm{Null}(\alpha) = S$*; moreover S is a Moishezon surface covered by a family of α-trivial curves, and $S \cdot R < 0$. In particular $R = \mathbb{R}^+[C_t]$ for the curves C_t of this family.*

Idea of proof. Perturb α by a small Kähler class and take the Boucksom–Zariski decomposition $\alpha - \epsilon\omega = \sum s_j S_j + P$. A component $S = S_1$ of the negative part carries $\alpha|_{S'}$ nef but not big; feeding the decomposition through adjunction shows $K_{S'}$ is not pseudo-effective, so the minimal resolution S' is uniruled, hence projective and Moishezon. A movable, α-trivial curve on S' is produced by Batyrev's decomposition of the nef curve class $\alpha|_{S'}$; being movable it covers S.

Proof. Let ω be Kähler. As α is big, $\alpha - \epsilon\omega$ is big for $0 < \epsilon \ll 1$, with Boucksom–Zariski decomposition $\alpha - \epsilon\omega = \sum s_j S_j + P$, P modified nef and big [?]. If $S \subset \mathrm{Null}(\alpha)$ is a surface and $S' \to S$ its minimal resolution, then $\alpha|_{S'}$ is nef with $(\alpha|_{S'})^2 = 0$, so S is a component of the negative part, say $S = S_1$.
Put $b = \mathrm{mult}_S(B)$. Since $\mathrm{mult}_S\big(B + \frac{1-b}{s_1} \sum s_j S_j\big) = 1$, adjunction on S' gives

$$\left(1 + \tfrac{1-b}{s_1}\right)\alpha\Big|_{S'} = K_{S'} + B_{S'} + \left(\beta + \tfrac{1-b}{s_1}(P + \epsilon\omega)\right)\Big|_{S'}, \qquad B_{S'} \geq 0.$$

The bracketed class is big (Kähler + pseudo-effective) while $\alpha|_{S'}$ is not big, so $K_{S'}$ is not pseudo-effective. Hence S' is a projective uniruled surface with $H^2(S', \mathcal{O}_{S'}) = 0$, and S is Moishezon.
Because $H^2(S', \mathcal{O}_{S'}) = 0$, the nef class $\alpha|_{S'}$ is represented by an $\mathbb{R}$-divisor, and as a nef curve it lies in $\overline{\mathrm{NM}_1}(S')$. Batyrev's theorem [?] gives $\alpha|_{S'} \equiv C_\epsilon + \sum_i \lambda_i M_i$ with M_i movable. If some $\alpha|_{S'} \cdot M_i = 0$, then M_i pushes to a movable α-trivial curve on S and we are done. Otherwise $\alpha|_{S'} \cdot M_i > 0$ for all i; then $0 = (\alpha|_{S'})^2 = \alpha|_{S'} \cdot (C_\epsilon + \sum \lambda_i M_i)$ forces $k = 0$ and $\alpha|_{S'} \equiv C_\epsilon$, whence $\alpha|_{S'} \cdot (K_{S'} + \epsilon A_{S'}) \geq 0$ and, letting $\epsilon \to 0$, $\alpha|_{S'} \cdot K_{S'} \geq 0$. Combined with the displayed adjunction identity and $(\alpha|_{S'})^2 = 0$, every term is forced to vanish; in particular $\alpha|_{S^\nu} \cdot \omega|_{S^\nu} = 0$ on the normalization, so $\alpha|_{S^\nu} \equiv 0$ — contradicting $\alpha|_{S'} \cdot M_i > 0$. Thus the covering family exists.
Finally, for a covering α-trivial curve C_t,

$$0 > -\epsilon\omega \cdot C_t = (\alpha - \epsilon\omega) \cdot C_t = \left(\sum s_j S_j + P\right) \cdot C_t \geq s_1\, S \cdot C_t,$$

using $S_j \cdot C_t \geq 0$ for $j \neq 1$. Hence $S \cdot C_t < 0$, so $S \cdot R < 0$. The same computation applied to any curve C with $[C] \in R$ gives $S \cdot C < 0$, forcing $S = \mathrm{Null}(\alpha)$ with no other component. ∎

2.3 Step 2: the nef reduction dichotomy

Let $S^\nu \to S$ be the normalization and $n(\alpha) := n(\alpha|_{S^\nu})$ the nef dimension of $\alpha|_{S^\nu}$. Since S is covered by α-trivial curves, $\alpha|_{S^\nu}$ is not big, so $0 \le n(\alpha) \le 1$. The nef reduction [?] is a morphism $\varphi\colon S^\nu \to T$ with:

- $n(\alpha) = 0$: $\alpha|_{S^\nu} \equiv 0$ and T is a point;
- $n(\alpha) = 1$: T is a smooth curve, $\alpha|_{S^\nu} = \varphi^*\gamma$, and a curve C is α-trivial iff it is vertical over T.

The whole construction below runs uniformly in $n(\alpha)$, the two cases differing only in which components of μ^*S end up contracted.

2.4 Step 3: a $\mathbb{Q}$-factorial minimal dlt modification

Put $b = \mathrm{mult}_S(B)$ and $\Delta = B + (1-b)S$, so $\lfloor \Delta \rfloor \supseteq S$ and $\alpha = [K_X + B + \beta]$.

Proposition 2.4 ([?, Claim 6.11])**.** *There is a projective bimeromorphic $\mu\colon X' \to X$ such that, with $\Delta' = \mu_*^{-1}\Delta + \mathrm{Ex}(\mu)$:*

(1) (X', Δ') is $\mathbb{Q}$-factorial and DLT;

(2) $K_{X'} + \Delta'$ is nef over X, so $\mu^*(K_X + \Delta) - (K_{X'} + \Delta') \ge 0$;

(3) $K_{X'} + \Delta' = \mu^*(K_X + B) + (1-b)S' + \sum a_j E_j$ with $S' = \mu_*^{-1}S$ and all $a_j > 0$.

Proof. Take a projective log resolution $\mu\colon X' \to X$ with $\Delta' = \mu_*^{-1}\Delta + \mathrm{Ex}(\mu)$; since $\mathrm{Ex}(\mu)$ is SNC, (X', Δ') is log smooth, hence DLT. Run the $(K_{X'} + \Delta')$-MMP over X (analytic BCHM [?]); it is automatically of general type over X as μ is bimeromorphic, and it preserves DLT and $\mathbb{Q}$-factoriality, giving (1). By termination we may assume $K_{X'} + \Delta'$ is nef over X; the negativity lemma then yields $\mu^*(K_X + \Delta) - (K_{X'} + \Delta') \ge 0$, which is (2). For (3): as (X, B) is klt, $\mu^*(K_X + B) = K_{X'} + \tilde{B} + \sum b_j E_j$ with $b_j < 1$; comparing with $\Delta' = (1-b)S' + \tilde{B} + \mathrm{Ex}(\mu)$ gives $\mathrm{Ex}(\mu) = \sum (b_j + a_j) E_j$, so $a_j = 1 - b_j > 0$. ∎

Condition (2) is the technical payload — it provides the relative nef/positivity later fed to the negativity lemma — while (3) records that the locus we intend to contract (S' and $\mathrm{Ex}(\mu)$) is retained with positive coefficient, hence sits inside $\lfloor \Delta' \rfloor$.

2.5 Step 4: the modification is trivial away from the null locus

Write $U = X \setminus S = X \setminus \mathrm{Null}(\alpha)$, $X'_U = \mu^{-1}(U)$, $\alpha' = \mu^*\alpha$.

Lemma 2.5 ([?, p. 36]). *(0)* $(X, \Delta|_U) = (U, B|_U)$ *is klt;*

1. $\mathrm{Ex}(\mu)$ *is divisorial and* $\mu|_{X'_U} : X'_U \to U$ *is an isomorphism;*
2. *if a curve* $C \not\subset X' \setminus X'_U$*, then* $\alpha' \cdot C > 0$*;*
3. *every* α'*-trivial step of the* $(K_{X'} + \Delta')$*-MMP involves only curves in* $X' \setminus X'_U$*.*

Proof. On U, $\Delta|_U = B|_U$ since $S = \mathrm{Null}(\alpha)$ is removed, so $(U, \Delta|_U)$ is klt. As X is $\mathbb{Q}$-factorial and X' has rational (DLT) singularities, $\mathrm{Ex}(\mu)$ is a divisor, so $\mu|_{X'_U}$ is an isomorphism. For (2): $\mu^*\alpha \cdot C = \alpha \cdot \mu_* C \geq 0$ by nefness, and if it were 0 then $\mu(C) \subset \mathrm{Null}(\alpha) = X \setminus U$, forcing $C \subset X' \setminus X'_U$; contrapositive is (2). Then (3) is immediate, since α'-trivial contracted curves have $\alpha' \cdot C = 0$ and hence lie in $X' \setminus X'_U$. ■

2.6 Step 5: running the α'-trivial MMP

Proposition 2.6 ([?, Claim 6.12]). *One can run an* α'*-trivial* $(K_{X'} + \Delta')$*-MMP* $X' = X^0 \dashrightarrow X^1 \dashrightarrow \cdots \dashrightarrow X^m$ *with* $\phi^i : X' \dashrightarrow X^i$ *such that:*

1. *each step is* α'*-trivial, so* $\alpha^i = \phi^i_* \alpha'$ *is nef;*
2. $\phi^i|_{X'_U}$ *is an isomorphism onto* X^i_U*;*
3. $(X^i, \Delta^i = \phi^i_* \Delta')$ *is DLT,* X^i *strongly* $\mathbb{Q}$*-factorial, with* $\lfloor \Delta^i \rfloor \subset W^i := X^i \setminus X^i_U$*;*
4. $K_{X^i} + \Delta^i - \Theta^i \equiv_{\alpha^i} 0$ *with* $\mathrm{Supp}(\Theta^i) = \lfloor \Delta^i \rfloor$*;*
5. *if* S^m *is a component of* $\lfloor \Delta^m \rfloor$*, then every* $(K_{S^m} + \Delta_{S^m})$*-negative extremal ray* R^m *is* α_{S^m}*-positive,* $\alpha_{S^m} \cdot R^m > 0$*.*

Idea of proof. An ordinary MMP need not exist here; the point is to run it α'*-trivially*, so that the only contractions performed are the ones supported on $\lfloor \Delta' \rfloor = \mu_*^{-1} S \cup \mathrm{Ex}(\mu)$. Each contraction is manufactured by finding a $(K_{S^m} + \Delta_{S^m})$-negative, α-trivial extremal ray *on a boundary surface* $\bar{S}^i$, contracting it there, and extending the surface contraction to X^i by the PLT theorem **??**. Termination is the 3-fold termination of [?]; condition (5) is the stopping criterion, signalling nothing α-trivial is left to contract on the NKLT locus.

Proof sketch. *Base case.* $\alpha' = \mu^*\alpha$ is nef. Since $(K_X + B) \cdot R < 0$ and $S \cdot R < 0$, there is $s > 0$ with $(K_X + B - sS) \cdot R = 0$, i.e. $K_X + B - sS \equiv_\alpha 0$; pulling back,

$$K_{X'} + \Delta' \equiv_{\alpha'} \Theta' := (1 - b)S' + \sum a_j E_j + s\mu^* S \geq 0, \qquad \mathrm{Supp}(\Theta') = \lfloor \Delta' \rfloor.$$

Inductive step. Suppose a component $S^i \subset \lfloor \Delta^i \rfloor$ carries a $(K_{S^i} + \Delta_{S^i})$-negative, α_{S^i}-trivial extremal ray $R^i = \mathbb{R}^+[\Sigma^i]$ (otherwise (5) holds and we stop). From (4), $\Theta^i \cdot \Sigma^i = (K_{X^i} + \Delta^i) \cdot \Sigma^i < 0$, so some component $\bar{S}^i$ of $\lfloor \Delta^i \rfloor$ has $\bar{S}^i \cdot \Sigma^i < 0$ and $\Sigma^i \subset \bar{S}^i$. On the surface $\bar{S}^i$, Σ^i lies in a $(K_{\bar{S}^i} + \Delta_{\bar{S}^i})$-negative, $\alpha_{\bar{S}^i}$-trivial extremal face; a spanning ray $\bar{\Sigma}^i$ has $\bar{S}^i \cdot \bar{\Sigma}^i < 0$, and its extremal contraction $\pi : \bar{S}^i \to V$ satisfies: $-\bar{S}^i$ is π-ample and

$-(K_{\bar{S}^i} + \Delta_{\bar{S}^i})$ is relatively Kähler (Picard number 1 over V). The pair $(X^i, \bar{S}^i + B^i)$ is PLT, so **??** supplies a contraction $p\colon X^i \to Z^i$ with $p|_{\bar{S}^i} = \pi$; if p is flipping the flip exists. Set $X^i \dashrightarrow X^{i+1}$. Termination [?] yields X^m with (5). Conditions (1)–(4) are preserved: α^i-triviality of p and injectivity of pullback of Bott–Chern classes under Fano-type contractions give $\alpha^i \equiv p^*\alpha_{Z^i}$ with α_{Z^i} nef, and $K_{X^i} + \Delta^i - \Theta^i \equiv_{\alpha^i} 0$ pushes/pulls through each step. ■

2.7 Step 6: controlling what is contracted, and $\Theta^m = 0$

The MMP $\phi^m\colon X' \dashrightarrow X^m$ is now shown to contract exactly μ^*S, so that downstairs $\phi\colon X \dashrightarrow X^m$ contracts precisely S.

Proposition 2.7 ([?, Claims 6.13–6.14]). *(1) If $n(\alpha) = 0$, then ϕ^m contracts μ^*S; hence $\phi\colon X \dashrightarrow X^m$ contracts S and no other divisor.*

(2) If $n(\alpha) = 1$, then ϕ^m contracts S' and every component of μ^*S dominating T; hence ϕ contracts S and may extract only divisors over S not dominating T.

In all cases $\Theta^m = 0$.

Idea of proof. Choose $F' \geq 0$ μ-exceptional with $-F'$ μ-ample and set $\Xi' = \mu^*S + \epsilon F'$. Two facts drive the proof. First, $-(\Xi' - t\alpha')$ restricts to a Kähler (ample over T) class on S' and on each E_j — because $-S|_{S^\nu} \equiv_\alpha \frac{1}{s}\beta|_{S^\nu}$ is relatively ample and α is nef. Second, the negativity lemma gives $p^*\Xi' - q^*\Xi^m \geq 0$ on a common resolution. If a component of μ^*S survived, a covering family C_t^m of α^m-trivial curves on it would satisfy, using (5) and $K_{X^m} + \Delta^m \equiv_{\alpha^m} \Theta^m$,

$$0 \geq (\Theta^m - \lambda\Xi^m)\cdot C_t^m = (K_{X^m} + \Delta^m - \lambda\Xi^m)\cdot C_t^m \geq -\lambda\Xi^m \cdot C_t^m \geq -\lambda\Xi' \cdot C_t > 0,$$

a contradiction. Hence every intended component is contracted, forcing $\Theta^m = 0$.

Proof that $\Theta^m = 0$. It suffices to show $q^*\Theta^m = 0$ on the common resolution $p\colon W \to X$, $q\colon W \to X^m$, since pullback of Bott–Chern classes under the contraction μ is injective. As $\Theta^m \geq 0$, by the negativity lemma we need $q^*\Theta^m$ effective, exceptional and nef over X. Exceptionality: by **??** both S' and $\mathrm{Ex}(\mu)$ are contracted, so $\mathrm{Supp}(\Theta^m) = \lfloor \Delta^m \rfloor$ meets no strict transform surviving over X. Nefness: take $C \subset W$ with $\mu_* p_* C = 0$. If $q_* C = 0$ then $q^*\Theta^m \cdot C = 0$. If $q_* C \neq 0$ then $\alpha^m \cdot q_* C = \alpha \cdot \mu_* p_* C = 0$, so $q_* C$ is α^m-trivial; if it lies in a component P of Θ^m, then by (5) $(K_{X^m} + \Delta^m)|_P$ is ≥ 0 on α^m-trivial curves, whence

$$q^*\Theta^m \cdot C = (K_{X^m} + \Delta^m)\cdot q_* C \geq 0.$$

Thus $q^*\Theta^m$ is nef over X, and $\Theta^m = 0$. ■

2.8 Step 7: the map descends to a morphism

Proposition 2.8 ([?]). *The bimeromorphic map $\phi\colon X \dashrightarrow X^m$ is a morphism $f\colon X \to Z := X^m$.*

Proof. Let $p\colon W \to X$, $q\colon W \to X^m$ resolve ϕ. By the rigidity lemma it suffices that every p-exceptional curve is q-exceptional. Pick a Kähler class ω^m on X^m. With $E_1, \ldots, E_r$ a basis of $H^{1,1}_{\mathrm{BC}}(W)/p^*H^{1,1}_{\mathrm{BC}}(X)$, write $p^*\omega \equiv q^*\omega^m + \sum e_i E_i$, and choose r with $(\omega + rS)\cdot R = 0$ (possible since $S \cdot R < 0$); then

$$p^*(\omega + rS) \equiv q^*\omega^m + Q, \qquad Q = \sum e_i E_i + rp^*S.$$

For C q-exceptional: if $p_*C = 0$ then $(q^*\omega^m + Q)\cdot C = (\omega + rS)\cdot p_*C = 0$; if $p_*C \neq 0$, then $p^*\alpha \cdot C = q^*\alpha^m \cdot C = 0$ shows $p_*C \in R$, so again $(q^*\omega^m + Q)\cdot C = (\omega + rS)\cdot p_*C = 0$. Thus Q is q-trivial. By **??** ($\Theta^m = 0$) ϕ contracts S and extracts no divisor, so each E_i and p^*S is q-exceptional, hence Q is q-exceptional. The negativity lemma forces $Q = 0$, i.e. $p^*(\omega + rS) \equiv q^*\omega^m$. If some p-exceptional C were not q-exceptional,

$$0 = C \cdot p^*(\omega + rS) = C \cdot q^*\omega^m = q_*C \cdot \omega^m > 0,$$

absurd. So $\phi = f$ is a morphism. ■

2.9 Step 8: base-point freeness and $\alpha = f^*\alpha_Z$

Proposition 2.9 ([?]). *$-(K_X + B)$ is f-ample, and there is a Kähler class α_Z on Z with $\alpha = f^*\alpha_Z$. This completes* **??**.

Proof. Since μ and ϕ^m are isomorphisms over U, so is f, and f contracts exactly the α-trivial curves, i.e. the curves in R. From $\alpha = [K_X + B + \beta]$ and $\alpha \equiv_Z 0$ we get $-(K_X + B) \equiv_Z \beta$, which is f-ample. Hence Z has rational singularities, and injectivity of f^* on Bott–Chern classes gives $\rho(X/Z) = 1$ and $\alpha \equiv f^*\alpha_Z$ for some α_Z.
It remains to see α_Z is Kähler; by Demailly–Păun [?] it suffices that $\alpha_Z^{\dim W} \cdot W > 0$ for every positive-dimensional $W \subset Z$. If $W \not\subset f(S)$, its strict transform $W' \not\subset S = \mathrm{Null}(\alpha)$, so $\alpha_Z^{\dim W} \cdot W = \alpha^{\dim W'} \cdot W' > 0$. If $W \subset f(S)$, then since R is divisorial $W = f(S)$ is a curve; S is projective (it is f^{-1} of a projective curve), so there is a curve $W' \subset S$ with $f(W') = W$ and $W' \notin R$, giving $\alpha \cdot W' > 0$ and $\alpha_Z \cdot W = \alpha \cdot W' > 0$ by the projection formula. Thus α_Z is Kähler. ■

3 The Höring–Peternell approach

The earlier proof, specific to threefolds, contracts the Moishezon surface $S = \mathrm{Null}(\alpha)$ by classical surface-and-bundle geometry rather than a lifted MMP. The starting data are the same as in **??**: a divisorial K_X-negative extremal ray R with $S = \mathrm{Null}(\alpha)$ covered by the curves of R. Everything again splits on $n(\alpha) = n(\alpha|_{S^\nu}) \in \{0, 1\}$.

3.1 Contraction to a point ($n(\alpha) = 0$)

Theorem 3.1 ([?, Cor. 7.7]). *If R is divisorial and $n(\alpha) = 0$, then S blows down to a point: there is a bimeromorphic morphism $\varphi\colon X \to Y$ to a normal compact threefold with $\dim \varphi(S) = 0$ and $\varphi|_{X\setminus S}$ an isomorphism.*

Proof. Since $n(\alpha) = 0$, $\nu^*\alpha \equiv 0$ on the normalization $\tilde{S}$, so $\alpha \cdot C = 0$ for every curve $C \subset S$, i.e. $\alpha|_S \equiv 0$ and every such $[C] \in R$. Let m be the Cartier index of S. By the surface structure of R [**?**], $-mS$ is strictly positive on R; and α is strictly positive on the rest of $\overline{\mathrm{NA}}(X)$ (the supporting-class property of R). Hence for small $\varepsilon > 0$, $\alpha - \varepsilon[mS]$ is strictly positive on $\overline{\mathrm{NA}}(X) \setminus \{0\}$, so by the Kleiman criterion for Kähler 3-folds [**?**] it is a Kähler class $[\omega]$. As $\alpha|_S \equiv 0$, the Cartier class $[-mS|_S]$ is represented by $\frac{1}{\varepsilon}\omega$, so $\mathcal{O}_S(-mS)$ is ample. Grauert's contractibility theorem then blows S down to a point. ∎

3.2 Contraction to a curve ($n(\alpha) = 1$)

Here the nef reduction is a genuine fibration and the surface is contracted along it.

Lemma 3.2 ([?, Lem. 7.8]). *If R is divisorial and $n(\alpha) = 1$, there is a fibration $f : S \to B$ onto a curve such that a curve $C \subset S$ is contracted iff $[C] \in R$; its general fibre is $\mathbb{P}^1$.*

Proof sketch. Let $\tilde{C}$ be a general fibre of the nef reduction $\tilde{f}\colon \tilde{S} \to \tilde{B}$ of $\nu^*\alpha$; then $\nu^*\alpha \cdot \tilde{C} = 0$, so $\alpha \cdot \nu(\tilde{C}) = 0$ and $C := \nu(\tilde{C})$ has $[C] \in R$, whence $K_X \cdot C < 0$. Because C moves in a positive-dimensional family (divisorial type), the Zariski-decomposition analysis of K_X-negative curves gives $K_S \cdot C < K_X \cdot C < 0$; choosing $\tilde{C}$ through smooth points, $K_S \cdot C \leq -2$ and $K_{\tilde{S}} \cdot \tilde{C} \leq K_S \cdot C \leq -2$. But $\tilde{C}$ is a fibre of $\tilde{f}$ with $\tilde{C}^2 = 0$, so $\deg K_{\tilde{C}} = K_{\tilde{S}} \cdot \tilde{C} \geq -2$; hence equality, $\tilde{C} = C \cong \mathbb{P}^1$ with $K_S \cdot C = -2$. Since $\mathcal{O}_S(C)$ is invertible ($C \subset S_{\mathrm{reg}}$) and $C^2 = 0$ with $K_{\tilde{S}} \cdot C = -2$, asymptotic Riemann–Roch on $\tilde{S}$ (transported to S through the normalization sequence, whose cokernel is supported off C) gives $\kappa(S, \mathcal{O}_S(C)) = 1$; a multiple of C defines the fibration $f : S \to B$. ∎

Theorem 3.3 ([?, Cor. 7.9], [?, Thm. 1.1]). *With $f : S \to B$ as in **??**, S contracts onto the curve: there is a bimeromorphic morphism $\varphi\colon X \to Y$ to a normal compact threefold with $\varphi|_S = f$ (up to Stein factorization) and $\varphi|_{X\setminus S}$ an isomorphism.*

Idea of proof. Off a finite set $Z \subset B$ the fibration is a $\mathbb{P}^1$-bundle $f^\circ\colon S^\circ \to B^\circ$ with $\mathcal{O}_S(-mS)$ f-ample, so the conormal bundle $N^*_{S^\circ/X^\circ} \cong \mathcal{O}_{S^\circ}(-mS)$ is f°-ample.

> Nakano's theorem contracts S° fibrewise to B°; a divisorial-extension theorem of Höring–Peternell then globalizes the contraction over the finitely many bad fibres.

Proof sketch. Replacing f by its Stein factorization we may assume $f_*\mathcal{O}_S = \mathcal{O}_B$. As X is $\mathbb{Q}$-factorial, S is $\mathbb{Q}$-Cartier of some index m, and since f contracts an extremal ray, $\mathcal{O}_S(-mS)$ is f-ample. Because X is terminal, $\mathrm{Sing}(X)$ is finite; setting $Z = f(\mathrm{Sing}\,X) \cup \mathrm{Sing}(B)$ and removing $f^{-1}(Z)$, the restriction $f^\circ\colon S^\circ \to B^\circ$ is a $\mathbb{P}^1$-bundle with $N^*_{S^\circ/X^\circ}$ f°-ample. By Nakano's theorem there is $\varphi^\circ\colon X^\circ \to Y^\circ$ realizing f° as the blow-up of Y° along B°. Finally, the three hypotheses of the Höring–Peternell divisorial-contraction theorem [?] hold — $\mathcal{O}_S(-mS)$ f-ample, the extension φ° over $X \setminus f^{-1}(Z)$, and the isomorphism $\mathcal{I}^k_{B\setminus Z} \to \varphi^\circ_* \mathcal{I}^k_{S\setminus f^{-1}(Z)}$ at the generic point of B — so the contraction extends to $\varphi\colon X \to Y$. ■

4 Comparison of the two approaches

Key takeaways

- Both proofs localize the whole problem on the Moishezon surface $S = \mathrm{Null}(\alpha)$ (**??**) and split on the nef dimension $n(\alpha|_{S^\nu}) \in \{0, 1\}$ — contraction to a point versus to a curve.
- **Höring–Peternell** contract S by hand: Kleiman + Grauert to a point, or a $\mathbb{P}^1$-bundle structure + Nakano + a divisorial-extension theorem to a curve. This uses surface classification, Riemann–Roch on $\tilde{S}$, and terminal 3-fold geometry throughout — it is intrinsically 3-dimensional.
- **Das–Hacon** replace all of this by a single mechanism: lift to a dlt model, run an α'-trivial MMP that surgers only the null locus, and feed each elementary contraction to the all-dimensional PLT contraction theorem **??**. Nothing in the logic is special to dimension three except the inputs it quotes (3-fold termination, the Moishezon-surface structure of $\mathrm{Null}(\alpha)$).
- Consequently the Das–Hacon method is the one that survives into higher dimension, while Höring–Peternell remains the more geometric and concrete account in dimension three.